

Paolo De Marinis

MSc in Mathematics | zk-STARKs & Cryptography | DLT & Technical Projects

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Professional Profile

MSc in Mathematics (106/110) with a thesis written in English on zk-STARK protocols. The thesis develops a largely self-contained theoretical reconstruction of zk-STARKs from first principles and includes a formal finite-state-machine $\rightarrow$ Computational Integrity statement $\rightarrow$ AIR derivation with proof of equivalence, a general AIR definition unifying formulations in the literature, and an optimization framework based on two preorders for valid and probabilistically sound extensions. It also reconstructs the STARK protocol through Reed-Solomon encoding, DEEP and FRI, with complexity and soundness analysis. My structured self-study in cryptography has covered modern cryptographic security foundations, symmetric cryptography, hash functions and MACs, core public-key topics including RSA and Diffie-Hellman, ECDSA and Schnorr digital signatures, and the foundations of Bitcoin, Ethereum and DLT. Continuing study includes additional public-key topics, PKI/TLS, zero-knowledge systems beyond zk-STARKs, advanced DLT topics, privacy-enhancing technologies and post-quantum cryptography. Direct programming and scientific-computing experience includes C, C++ through university coursework, and MATLAB; Python experience is project-based. Linux user since 2012.

Education

MSc in Mathematics (LM-40) - 106/110

2023-2025

University of Insubria, Italy

Thesis: *Fast Fourier Transform in zk-STARK Protocols: Interpolation, Encoding and Proximity Proofs*. Supervisors: Marco Donatelli and Alberto Trombetta.

- **Scope and method:** largely self-contained theoretical reconstruction from first principles, intended to bridge a didactic gap in the literature.
- **Formal FSM-based AIR derivation:** finite-state machine $\rightarrow$ Computational Integrity statement $\rightarrow$ AIR, with proof of logical equivalence between AIR satisfiability and CI-statement validity.
- **General AIR definition:** unifying model combining local, non-local and state constraints with distinct evaluation domains.
- **Optimization framework:** two preorder relations for valid and probabilistically sound extensions, enabling transformations to be classified, compared and composed through transitivity.
- **Analysis and reinterpretation:** trace padding, permutation arguments and preprocessing; formalized constraint unrolling and random reaggregation; combined these techniques in a theoretical Randomized AIR with Preprocessing for an ISA-based zk-VM.
- **Protocol reconstruction and analysis:** Reed-Solomon encoding, DEEP and FRI, with complexity and FRI soundness analysis via the Correlated Agreement Theorem in the unique-decoding regime. The thesis also covers the hash-based commitment structure and Merkle trees used within the zk-STARK construction.

Technical Skills

- **Cryptography foundations - studied topics:** foundations and security models of modern cryptography; symmetric cryptography; hash functions and MACs; core public-key topics including RSA and Diffie-Hellman; ECDSA and Schnorr digital signatures, including their algebraic foundations and security considerations; and the foundations of Bitcoin, Ethereum and DLT, including major consensus mechanisms.
- **Continuing study:** additional public-key cryptography topics, PKI/TLS, zero-knowledge systems beyond zk-STARKs, advanced DLT topics, privacy-enhancing technologies and post-quantum cryptography.
- **Blockchain practice:** Bitcoin/Ethereum self-custody with Ledger and MetaMask, dApps and DeFi, wallet-based transaction signing, smart-contract interaction and token approvals. Initial experimentation with Solidity, without substantial project-based Solidity development.
- **Programming/scientific computing:** direct C and MATLAB experience. University C++ work included procedural, C-style programming in the *Programming and Computing Laboratory (Laboratorio di Programmazione e Calcolo)*; object-oriented C++ was limited to a single program for the pass/fail *Programming Languages (Linguaggi di Programmazione)* exam. Algorithms and data structures were covered in *General Computer Science (Informatica Generale)*. GNU Octave was used directly in the *Foundations of Numerical Analysis (Istituzioni di Analisi Numerica)* exam at the University of Pisa; Maple was learned independently and used for orbital-transfer and rendezvous calculations presented in the *Space Mechanics (Meccanica Spaziale)* examination seminar.
- **Python project work:** pipeline structure and design, code reading/review, targeted changes, testing and validation for a reversible pipeline mapping 2D points to Hilbert indices, ordering the indices and applying delta encoding to the ordered sequence, with inverse decoding from deltas to indices and from indices back to points; implemented with `hilbertcurve`, with targeted ChatGPT support for some functions.
- **Technical environment and tools:** Linux daily since 2012; Bash, Git/GitHub and GitHub Actions; advanced LaTeX for academic and technical documentation; project-specific experience with ACPI/AML diagnostics and validation, initramfs and Limine through OMEN ACPI Toolkit.
- **Agentic workflows:** personal responsibility for initial objectives and constraints, context preparation, diff review, testing, validation and final acceptance; agents used for technical formalization, implementation, automated tests and documentation.

Selected Projects

RIVES cartridges - C games and procedural geometry

2024; GitHub repositories 2026

Original production and Jam (Oct–Dec 2024): conceived, developed and published *Bomb Flip* first and then *Slither Slide* in C using RIVES ([riv.h](#)); used Ethereum through a wallet for platform transactions. For *Bomb Flip*, inspired by Voltorb Flip but not a clone, I designed the specific rules and wrote most original gameplay code; Cursor mainly supported animation and implementation details. For *Slither Slide*, I designed mechanics, worked on parts of the code, integrated/validated the result and selected/configured the closed uniform cubic B-spline; assistance was greater. *Slither Slide* placed 2nd out of 7 entries in RIVES Jam #3. GitHub repository phase (Aug 2026): I created and now maintain the public *Bomb Flip* (v1.0.0) and *Slither Slide* (v1.0.0) repositories from the available post-jam versions, not exact 2024 source snapshots. I integrated provenance, documentation, tests, CI, cartridges and checksums; Codex assisted with maintenance, refactoring, tests and documentation, which I reviewed and validated.

OMEN ACPI Toolkit - Linux/ACPI project

Jul 2026-present (maintenance)

[Repository](#) | [v2.4.0 release](#) (14 Aug 2026); unaffiliated with HP. Diagnosed incomplete S5 shutdown and a distinct WQBZ ACPI error on the sole physically validated configuration: HP OMEN MAX 16-ap0006sl, board 8E35, BIOS F.13. I own hardware evidence, objectives/constraints, safety decisions, review, log/regression analysis, real-hardware validation and releases; Codex made substantial contributions to Bash/Python implementation, tests and documentation. v2.4.0 improves packaging and release engineering; ACPI transformers, verifiers, patches and fixtures are unchanged from the v2.3.0 validated baseline. No independent specialist audit or non-reference hardware validation is claimed.

Geospatial data compression - algorithmic contribution

2023

Voluntary, unpaid scientific collaboration; non-author contribution acknowledged in [Liverotti et al. \(2025\)](#). Contributed to the design and Python implementation of a reversible pipeline for 2D data: mapping points to Hilbert indices, ordering by those indices, applying delta encoding to the ordered sequence, decoding the deltas, and reconstructing the original points through the inverse Hilbert map. Personally verified round-trip reversibility on real Earth Observation data. The implementation used [hilbertcurve](#), with targeted ChatGPT support for some functions. I proposed Huffman coding for brightness values and contributed to the triangulation-reconstruction idea, without implementing those parts myself.

Blockchain & Community Background

Practical interest in blockchain began in 2016 and resumed through structured self-study and hands-on use from 2020. Experience includes Bitcoin and Ethereum self-custody, MetaMask, dApps, DeFi and smart-contract interaction, together with initial experimentation in Solidity. In 2022, the Cartesi Foundation published the community profile [Why a School Teacher Chose to Hold CTSI](#), covering my path from Ethereum and DeFi to the scalability problem and interest in Cartesi. The profile reflects community participation and personal study, not employment by Cartesi.

Additional Education

Master's studies in Mathematics - 75 ECTS completed before transfer

2016-2023

University of Pisa, Italy

Completed 75 ECTS through October 2020; enrollment was subsequently maintained during a study pause before transfer to the University of Insubria in 2023. Of the 75 ECTS, 71 were recognized on transfer. Coursework included *Foundations of Numerical Analysis*, *Scientific Computing (Calcolo Scientifico)*, *Solar System Dynamics (Dinamica del Sistema Solare)*, *Orbital Determination (Determinazione Orbitale)*, *Celestial Mechanics (Meccanica Celeste)*, *Space Mechanics and Planetary Systems (Sistemi Planetari)*. University coursework included direct use of GNU Octave in the *Foundations of Numerical Analysis* exam at Pisa; use of independently learned Maple for orbital-transfer and rendezvous calculations presented in the *Space Mechanics* examination seminar; preparation, for the *Orbital Determination* exam, of an independent 47-slide seminar on Gauss, Laplace and Moschetti methods for preliminary orbit determination; and an individual study of the Yarkovsky effect on asteroids for the *Planetary Systems* exam.

BSc in Mathematics (L-35)

2012-2016

Sapienza University of Rome, Italy

Coursework included procedural, C-style programming in C++ in the *Programming and Computing Laboratory*, algorithms and data structures in *General Computer Science*, and MATLAB, used in the undergraduate *Numerical Analysis (Analisi Numerica)* exam at Sapienza and in the BSc thesis. Thesis on Brouwer's fixed-point theorem and periodic solutions of ODEs.

Professional Experience

Computer Science Teacher (A-41) & Digital Trainer

Sep 2023-Aug 2024

IIS Leonardo da Vinci and Istituto Professionale Crotto Caurga, Italy

Taught computer science across technical and vocational programs, including C/C++ programming; served on the digital team and delivered Google Workspace/EIPASS-related training.

Mathematics Teacher

Sep 2022-Aug 2023

IIS Agrario Antonio della Lucia, Feltre, Italy

Taught mathematics across technical, vocational and professional-training programs.

Languages

Italian: native. English: B2; used to write the MSc thesis and technical documentation and to read scientific literature.